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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

AD1103-1 – SOFTWARE ENGINEERING

(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the primary objective of software engineering?**
A) To write code quickly
B) To produce software that is maintainable, reliable, and efficient
C) To focus only on hardware development
D) To eliminate all software bugs
2. **Which of the following is NOT a fundamental software engineering activity?**
A) Software specification
B) Software design and implementation
C) Software evolution
D) Hardware manufacturing
3. **Which of the following is an example of generic software?**
A) A company's payroll system
B) A traffic control system
C) A word processor like Microsoft Word
D) A custom hospital management system
4. **Which of these software product attributes is crucial for safety-critical systems?**
A) Maintainability
B) Dependability and security
C) Cost-effectiveness
D) User-friendliness
5. **Agile Software Development is based on:**
A) Incremental Development
B) Iterative Development
C) Linear Development
D) Both Incremental and Iterative Development
6. **Which of the following is NOT a phase in the waterfall model?**
A) System and software design
B) Implementation and unit testing
C) Marketing and advertising
D) Integration and system testing
7. **Which software model is best suited for large, complex systems?**
A) Waterfall model
B) Agile development
C) Spiral model
D) Throwaway prototyping
8. **What is the main challenge of developing safety-critical systems?**
A) Ensuring the system is reliable and free of critical failures
B) Reducing costs
C) Making the system open-source
D) Writing the code in Python
9. **UML is primarily used for:**
A) Writing software code
B) Representing system architecture using graphical models
C) Debugging and testing software
D) Developing new programming languages
10. **What is a key characteristic of system models?**
A) They include all details of the system
B) They are exact representations of the system
C) They focus on abstraction and simplification
D) They do not support documentation

11. **Sequence diagrams are primarily used for:**
 A) Showing how objects interact over time
 B) Defining system constraints
 C) Representing database schemas
 D) Mapping system performance metrics
12. **A state diagram in UML is used to model:**
 A) Data flow in the system
 B) The dynamic behavior of a system
 C) The physical architecture of the system
 D) User interfaces
13. **What is the purpose of aggregation in class diagrams?**
 A) It represents object composition relationships
 B) It shows control flow between system states
 C) It defines external dependencies
 D) It models user interface components
14. **The role of interaction models in system modeling is to:**
 A) Define how system components interact
 B) Describe system components' static structure
 C) Provide database relationships
 D) Explain hardware configurations
15. **The advantage of model - driven engineering is:**
 A) It reduces human involvement in coding
 B) It allows system implementation to be generated from models
 C) It makes software testing unnecessary
 D) It does not require system models
16. **What is the key difference between data-driven and event-driven modeling?**
 A) Data-driven models focus on event handling
 B) Event-driven models emphasize processing triggered by external stimuli
 C) Data-driven models do not process information
 D) Event-driven models are not dynamic
17. **Which of the following is a reason why software project management differs from other types of project management?**
 A) Software projects are always easy to estimate
 B) Software projects are always identical
 C) Software processes are variable and organization-specific
 D) Software projects never experience failure
18. **What is the primary goal of risk management in software projects?**
 A) To avoid any changes in software requirements
 B) To identify, analyze, and mitigate risks affecting project success
 C) To increase project costs
 D) To ensure that project risks are ignored
19. **Business risks in software projects include:**
 A) Unavailability of hardware
 B) Competitors launching a new product before your software is completed
 C) A delay in system testing
 D) Bugs in the software code
20. **What is the role of a project plan in software engineering?**
 A) To outline project scope, schedule, and risks
 B) To remove the need for risk management
 C) To ensure software development follows an unpredictable path
 D) To avoid planning activities

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the structure of a requirements document in requirement engineering.	8	L*2	1	1
	b) Explain the various checks involved in requirement validation in detail.	5	L2	1	1
	c) Explain the different types of interviews conducted during the requirement collection process in software development.	3	L2	2	1
2.	a) Briefly, discuss various phases of software development life cycle.	8	L2	1	1
	b) Briefly explain the essential attributes of good software. List out the fundamental software engineering activities.	8	L2	2	1
3.	a) Explain the Waterfall model. What are the advantages and disadvantages?	8	L2	1	1

- b) Explain the working of the Scrum framework in software development. Discuss the key activities involved in the Scrum process.

8 L2 3 1

Unit – II

4. a) Briefly explain the System Testing.
b) Describe the five key UML diagram types used in system modeling with their purposes.

8 L2 4 1

8 L3 3 1

5. a) Briefly explain the concept of system models.
b) Explain the architecture and working of language processing system in architectural design.

8 L2 4 1

8 L2 4 1

6. a) Briefly explain any four different test strategies used for conventional software development.
b) Explain the Software Architecture of an ATM System used in Transaction Processing Systems.

8 L2 3 1

8 L2 3 1

Unit – III

7. a) Explain the concept of project scheduling and its significance in project management.
b) Explain the concept of configuration management in software development projects.

8 L2 5 1

8 L2 5 1

8. a) List and explain the key stages in component measurement process.
b) Describe various risk identification in project management.

8 L2 5 1

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
